

Quiz 2

1. The large batch production is more efficient in:
 - Classical manufacturing systems
 - Modern manufacturing systems

Answer:

Classical manufacturing systems

2. What type of MS is used for producing office buildings?
 - Line
 - Project
 - Job shop
 - Continuous
 - Repetitive

Answer:

Project

3. Computer Aided Engineering (CAE) is used for:
 - Financial accounting
 - Process planning
 - Simulation and analysis of processes
 - Production control

Answer:

Simulation and analysis of processes

4. A continuous manufacturing system can be used if:
 - Product demand cannot be forecasted
 - Manufacturing lead time is smaller than the lead time expected by the customer
 - Product demand can be forecasted
 - Manufacturing lead time is greater than the lead time expected by the customer

Answer:

Product demand can be forecasted

Manufacturing lead time is greater than the lead time expected by the customer

5. Assembling a product requires 100 minutes when it is done for the first time by a trained worker. The process will be repeated to make 100 products. If the learning rate is 85%, what will be the delivery time of the 100th product? ($m=-0.23$, $100^m=0.35$).
 - 3.5
 - 35
 - 8.5

- 2.3

Answer: 35

6. Which of the following statements is/are true?
- The smaller the space - manufacturing ratio, the more efficient the production is.
 - The larger the space - manufacturing ratio, the more efficient the production is.
 - The larger the manufacturing - space ratio, the more efficient the production is.
 - The smaller the manufacturing - space ratio, the more efficient the production is.

Answer:

The larger the manufacturing - space ratio the more efficient the production is.

The smaller the space - manufacturing ratio, the more efficient the production is.

7. A line manufacturing system can be used if:
- Manufacturing lead time is greater than the lead time expected by the customer.
 - The product has many different options or models.
 - The product has only one model.
 - Manufacturing lead time is smaller than the lead time expected by the customer.

Answer:

Manufacturing lead time is greater than the lead time expected by the customer

The product has many different options or models.

8. The longest manufacturing time is found in the production system of type:
- Project
 - Line
 - Job shop
 - Repetitive
 - Continuous

Answer:

Project

9. A two stations production system has:
- Variable routing
 - No routing
 - Fixed routing
 - Mixed routing

Answer:

Fixed routing

10. Early detection of faults may reduce significantly the production costs.

- False
- True

Answer:
True

Quiz 3

1. Which of the following factors characterizes a CIM system?
 - Setup time
 - Flexibility
 - Quality
 - Inventory
 - Space-manufacturing ratio

Correct answer:

Setup time

Quality

Inventory

Flexibility

2. Utilization of a facility depends on:
 - Actual quantity produced during a given period
 - Production capacity for a given period
 - Mean time to repair
 - The level of integration

Correct answer:

Production capacity for a given period

Actual quantity produced during a given period

3. According to J.T. Black, the price of parts and materials is 50% of the selling price.
 - True
 - False

Correct answer: False

4. Computer-Aided Process Planning:
 - Decrease planning time
 - Increase planning time
 - Decrease human intervention
 - Increase flexibility

Correct answer:
Decrease planning time
Decrease human intervention
Increase flexibility

5. What is the best manufacturing strategy for a new type of product?
- Make to order
 - Make to stock
 - Assemble to order
 - Engineer to order

Correct answer: Engineer to order

6. Enterprise resource planning (ERP) is not efficient in CIM.
- False
 - True

Correct answer: False

7. What is the meaning of the ratio production capacity / quantity produced?
- It has no meaning
 - Mean time between failures
 - Availability
 - Utilization
 - Average manufacturing lead time

Correct answer: It has no meaning

8. What type of MS is used for producing toothpaste?
- Project
 - Job shop
 - Repetitive
 - Continuous
 - Line

Correct answer: Continuous

9. An operation that transforms the raw material from one state to another, the latter one being closer to the finished part or product is:
- A processing operation
 - A transport operation
 - A storage operation
 - An assembly operation

Correct answer:
A processing operation

10. A factory produces mobile phones and tablet PCs. On the line that produces mobile phones 5 models with minor differences can be made. On the line that produces tablet PCs, 5 models with major differences are produced. What is the total amount of products made by the company, if 100 phones of each model and 100 tablets PC of each model are produced?

- 1000
- 210
- 2500
- 5000

Correct answer: 1000

Quiz 5

1. In CIM, manufacturing batch size reduction leads to increased production costs.
- False
 - True

Answer: False

2. The setup time must be considered when determining the operation cycle time for job shop production.
- True
 - False

Answer: True

3. Each side of a one-aisle ASRS contains 10 storage compartments in the length direction and 10 compartments vertically. All storage compartments will be the same size to accommodate standard size pallets of dimensions: $x = 1$ m, $y = 1$ m. The height of a unit load is $z = 1$ m. What is the storage capacity of the ASRS?
- 10
 - 20
 - 200
 - 100

Answer: 200 (10 compartments x 10 compartments x 2 sides)

4. Each aisle of a four-aisle ASRS contains 10 storage compartments in the length direction and 10 compartments vertically. All storage compartments will be the same size to accommodate standard size pallets of dimensions: $x = 1$ m, $y = 1$ m. The height of a unit load is $z = 1$ m and the allowances are $a = 0.1$ m, $b = 0.1$ m, and $c = 0.1$ m. What is the width of the ASRS?
- 13.2

- 6.6
- 3.3
- 9.9

Answer: 13.2 (3 aisles x (1 + 0.1) m)

5. Which of the following statements is / are true?
- The throughput of an ASRS increase if transactions are performed in a dual command cycle.
 - The throughput of an ASRS decrease if transactions are performed in a dual command cycle.
 - The throughput of an ASRS increase if transactions are performed in a single command cycle.

Answer:

The throughput of an ASRS increase if transactions are performed in a dual command cycle.

6. Dedicated storage strategy is more appropriate when:
- Items are stored in part number sequence
 - Items are stored according to activity level
 - Items are stored according to their activity-to-space ratio

Answer:

Items are stored in part number sequence

Items are stored according to activity level

Items are stored according to their activity-to-space ratio

7. Which of the following statements is / are true?
- Higher throughput can be achieved when a dedicated storage strategy is implemented.
 - Less total space is required in a storage system that uses dedicated storage.
 - Higher throughput can be achieved when a random storage strategy is implemented.
 - Less total space is required in a storage system that uses randomized storage.

Answer:

Higher throughput can be achieved when a dedicated storage strategy is implemented

Less total space is required in a storage system that uses randomized storage

8. The handling of materials must be performed:
- In a timely manner
 - Inaccurately
 - At low cost
 - With damage to the materials

- Efficiently
- Safely

Answer:

In a timely manner

At low cost

Efficiently

Safely

9. What changes can be made to increase or decrease production capacity over short term?
- Change the number of shifts per week
 - Increase the number of work centers in the shop
 - Change the number of hours worked per shift
 - Increase the production rate, by making improvements in methods or process technology

Answer:

Change the number of shifts per week

Change the number of hours worked per shift

10. What is the meaning of the ratio quantity produced / production capacity?
- Mean time between failures
 - Availability
 - Average manufacturing lead time
 - Utilization
 - It has no meaning

Answer: Utilization

Quiz 6

1. Which of the following statements is / are true?
- The interpretation of each symbol from a hierarchical sequence depends on the previous symbol.
 - The interpretation of each symbol from a polycode sequence does not depend on the previous symbol.
 - The interpretation of each symbol from a monocode sequence does not depend on the previous symbol.
 - The interpretation of each symbol from a polycode sequence depends on the previous symbol.
 - The interpretation of each symbol from a chain-type sequence does not depend on the previous symbol.

Answer:

The interpretation of each symbol from a hierarchical sequence depends on the previous symbol

The interpretation of each symbol from a polycode sequence does not depend on the previous symbol

The interpretation of each symbol from a chain-type sequence does not depend on the previous symbol

2. When the work flow in the cell is analyzed:
- Percentage of in-sequence moves (PIM) should be as high as possible
 - Percentage of backtracking moves (PBM) should be as high as possible
 - Percentage of in-sequence moves (PIM) should be as low as possible
 - Percentage of backtracking moves (PBM) should be as low as possible

Answer:

Percentage of in-sequence moves (PIM) should be as high as possible

Percentage of backtracking moves (PBM) should be as low as possible

3. What is the number of levels required to represent the characteristics of a product, if a 4 digits monocode structure is considered?
- 3
 - 4
 - 5
 - There is no tree representation

Answer: 4

4. The costs associated with material handling must be:
- As small as possible
 - As large as possible
 - Does not matter

Answer: As small as possible

5. Which encoding structure provides the lowest information density?
- Hybrid
 - Polycode
 - Monocode
 - None of the above

Answer: Polycode

6. Which of the following are specific to project-type production systems?
- Simple drawings
 - Complex drawings
 - The customer is identified after production starts

- Cheap products
- Simple products
- Complex products
- Expensive products
- The customer is identified before production starts

Answer:

Complex drawings

Complex products

Expensive products

The customer is identified before production starts

7. Which of the following associations is / are correct?
- Production equipment $\Leftrightarrow$ machine groups
 - Similar parts $\Leftrightarrow$ part cells
 - Production equipment $\Leftrightarrow$ machine cells
 - Similar parts $\Leftrightarrow$ part families

Answer:

Production equipment $\Leftrightarrow$ machine groups

Production equipment $\Leftrightarrow$ machine cells

Similar parts $\Leftrightarrow$ part families

8. The complexity of a product is given by:
- The number of components
 - The number of operations needed for manufacturing

Answer:

The number of components

The number of operations needed for manufacturing

9. Random storage is used when products are not stored according to the level of activity.
- False
 - True

Answer: True

10. The use of group technology:
- Increases the setup time
 - Reduces the manufacturing lead time
 - Reduces the number of operations in the production process
 - Reduces the work-in-process
 - Increases the number of operations in the production process

Answer:

Reduces the manufacturing lead time

Reduces the number of operations in the production process

Reduces the work-in-process

Quiz 7

1. To be flexible a manufacturing system must:
 - Identify different part or product styles
 - Quick changeover the physical setup
 - Quick changeover the operating instructions

Answer:

- **Identify different part or product styles**
 - **Quick changeover the physical setup**
 - **Quick changeover the operating instructions**
2. Which of the following measures are used to evaluate the performance of a storage system?
 - Density
 - Storage capacity
 - Utilization
 - Reliability
 - Throughput
 - Accessibility

Answer:

- **Density**
 - **Storage capacity**
 - **Utilization**
 - **Reliability**
 - **Throughput**
 - **Accessibility**
3. A dedicated manufacturer of assemblies produces also the components necessary for their products.
 - True
 - False

Answer: False

4. Which of the following vehicle guidance technologies is the cheapest?
 - Paint stripes

- Imbedded guide wires
- Self guided vehicles

Answer: Paint stripes

5. Parts can be classified according to:
- Design and manufacturing attributes
 - Manufacturing attributes
 - Design attributes

Answer:

- **Design and manufacturing attributes**
 - **Manufacturing attributes**
 - **Design attributes**
6. Which of the following statements is / are true?
- When flexibility increases costs decrease.
 - When flexibility increases productivity decrease.
 - When flexibility increases productivity increase.
 - When flexibility increases costs increase.

Answer:

- **When flexibility increases productivity decrease.**
 - **When flexibility increases costs increase.**
7. The bottleneck station is:
- The station that gives the maximum production rate of the whole production system.
 - The station with the highest utilization.
 - The station that gives the minimum production rate of the whole production system.
 - The station with the lowest utilization.

Answer:

- **The station that gives the maximum production rate of the whole production system.**
 - **The station with the highest utilization.**
8. Which of the following is / are related to the part variety test?
- Can the system process different part styles in a non batch mode?
 - Can the system accept changes in production scheduling?
 - Can new part designs be introduced into the existing product mix with relative ease?
 - Can the system recover from equipment malfunctions without complete disruption of production?

Answer:

- **Can the system process different part styles in a non batch mode?**
9. Which of the following are specific to job shop - type production systems?
- Intensive routing
 - Intensive scheduling
 - High volume
 - Noncomplex products

Answer:

- **Intensive routing**
 - **Intensive scheduling**
 - **Noncomplex products**
10. A flexible manufacturing system consists of:
- Four or more processing workstations connected mechanically by a common part handling system and electronically by a distributed computer system.
 - Two or three processing workstations plus a handling system.
 - One CNC machining center combined with a parts storage system.

Answer: Four or more processing workstations connected mechanically by a common part handling system and electronically by a distributed computer system.

Quiz 8

1. Recirculating conveyors are the most efficient transport systems.
- False
 - True

Answer: False

2. Which of the following are benefits of using an FMS?
- Lower manufacturing lead times
 - Greater responsiveness to change
 - More machines required
 - Increased machine utilization
 - Increased inventory requirements

Answer:

- **Lower manufacturing lead times**
 - **Greater responsiveness to change**
 - **Increased machine utilization**
3. Processes planning is necessary for:
- Part produced

- Assembly

Answer:

- **Part produced**
- **Assembly**

4. The type of assembly method used for a given product depends on:
- Anticipated production quantities
 - Time required for each processing operation
 - Number of processing operations
 - Complexity of the assembled product

Answer:

- **Anticipated production quantities**
- **Complexity of the assembled product**

5. Group technology is most appropriately applied under the following conditions:
- The parts can not be grouped into part families.
 - The plant currently uses no batch production and a process type layout.
 - The plant currently uses traditional batch production and a process type layout.
 - The parts can be grouped into part families.

Answer:

- **The plant currently uses traditional batch production and a process type layout.**
- **The parts can be grouped into part families.**

6. The variety and quantity of goods are in a relationship of direct proportionality.
- False
 - True

Answer: False

7. What is the right order of the general steps required for the management activities of an investment project? (See attached image for steps)

- A. the equipment is installed and debugged;**
- B. vendor progress in building the equipment is monitored;**
- C. proposal to justify the investment is prepared;**
- D. management approvals are granted for the investment;**
- E. vendor quotations are solicited;**
- F. training of operators;**
- G. responsibility for running the equipment is turned over to the operating department.**
- H. order is placed to the winning vendor;**
- I. any special tooling and supplies are ordered;**

- E, B, D, G, C, F, A, H, I
- C, D, E, H, B, I, A, F, G
- C, A, B, E, D, F, H, I, G
- A, B, C, D, E, F, G, H, I
- D, E, B, C, G, A, F, H, I

Answer: C, D, E, H, B, I, A, F, G

8. What type(s) of design is/are not related to the concurrent engineering?
- Design for selling
 - Design for quality
 - Design for life cycle
 - Design for manufacturing and assembly
 - Design for cost

Answer: Design for selling

9. Which of the following are related to the use of CAPP?
- Productivity decreases
 - Lead time decreases
 - Lead time increases
 - Processes are standardized

Answer:

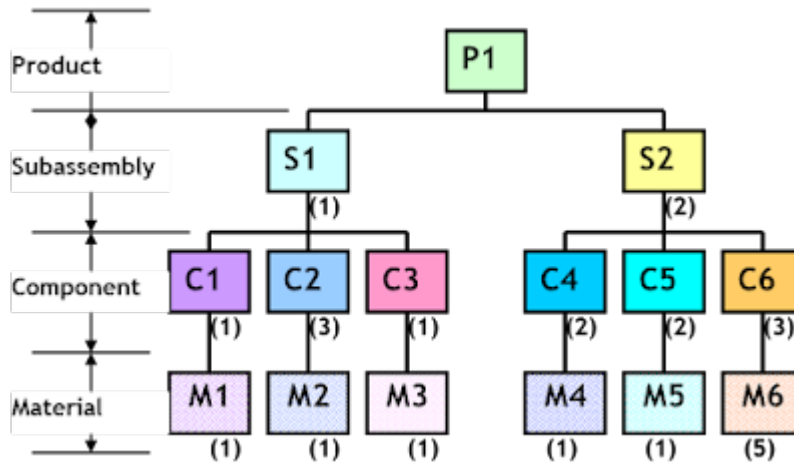
- **Lead time decreases**
- **Processes are standardized**

10. Concurrent engineering:
- Decreases the manufacturing lead time.
 - Increases the manufacturing lead time.
 - Does not affect the manufacturing lead time.

Answer: Decreases the manufacturing lead time

Quiz 9

1. The figure below shows the possible structure of a product (P1). How many units of material M5 are required for a batch of 10 products P1?



- 40
- 1
- 50
- 10
- 20

Answer: 40 (1 M5 * 2 C5 * 2 S2 * 10 P1)

2. The design of a material handling system does not depend on the type of manufacturing system where it is embedded.
 - False
 - True

Answer: False

3. The chase production strategy involves:
 - Cost of overtime
 - Cost of carrying the excess inventory created
 - Constant workforce
 - High cost of hiring and firing employees
 - No excess inventory is carried

Answer:

- **Cost of overtime**
 - **High cost of hiring and firing employees**
 - **No excess inventory is carried**
4. Which of the following statements is / are true?
 - The chase production strategy requires that the production in each period equals the planned production for that period.
 - The level production strategy requires that the production in each period equals the monthly average production calculated from the total production value for the year.

- The level production strategy requires that the production in each period equals the planned production for that period.
- The chase production strategy requires that the production in each period equals the monthly average production calculated from the total production value for the year.

Answer:

- **The chase production strategy requires that the production in each period equals the planned production for that period.**
- **The level production strategy requires that the production in each period equals the monthly average production calculated from the total production value for the year.**

5. Capacity adjustments for the short term include:

- Number of work shifts
- New plant construction
- Labor hours
- New equipment investments
- Temporary workers

Answer:

- **Number of work shifts**
- **Labor hours**
- **Temporary workers**

6. Designing for manufacturing and assembly involves:

- The use of special parts across product lines
- The use of modular design
- Minimizing the number of components
- The use of standard commercially available components
- Maximizing the flexible components

Answer:

- **The use of modular design**
- **Minimizing the number of components**
- **The use of standard commercially available components**

7. Which of the following statements is / are true?

- Less total space is required in a storage system that uses dedicated storage.
- Usually in a storage system the accessibility increases as the density increases.
- Less total space is required in a storage system that uses randomized storage.
- Higher throughput can be achieved when a random storage strategy is implemented based on activity level.

Answer:

- **Less total space is required in a storage system that uses randomized storage.**

8. The use of group technology decreases the number of operations in the production process.
- False
 - True

Answer: True

9. Production planning is concerned with:
- When the products should be completed
 - How many of each model
 - When to recall the products for service
 - Deciding what the selling price is
 - Deciding which products to make

Answer:

- **When the products should be completed**
- **How many of each model**
- **Deciding which products to make**

10. Which of the following statements is / are true?
- Aggregate production planning is used to convert master production schedule into an aggregate production plan.
 - Capacity planning refers to determining the quantity of products to be produced.
 - Master production planning is used to convert the aggregate production plan into a master production schedule.

Answer:

- **Master production planning is used to convert the aggregate production plan into a master production schedule.**

Quiz 10

1. What type(s) of card(s) is/are used in Kanban systems?
- For the authorization of the production
 - For the authorization of the delivery
 - For the authorization of the quality control
 - For the authorization of the planning
 - Only one type used for both authorization and delivery of the production

Answer:

- **For the authorization of the production**
- **For the authorization of the delivery**

2. Which of the following statements is / are correct regarding manufacturing resource planning (MRP II)?

- Inventories increase
- Inventories decrease
- Better collaboration with suppliers
- Better collaboration with customers

Answer:

- **Inventories decrease**
- **Better collaboration with suppliers**
- **Better collaboration with customers**

3. Aggregate production planning deals with the allocation of human resources.

- False
- True

Answer: False

4. For the chain-type structure, the number of digits of the code is determined by the number of levels in the hierarchy.

- False
- True

Answer: False

5. Which of the following questions are considered in the decision to make or buy a product?

- What is the total production quantity?
- What is the anticipated product life?
- Is the process available in-house?
- How do part costs compare?

Answer:

- **What is the total production quantity?**
- **What is the anticipated product life?**
- **Is the process available in-house?**
- **How do part costs compare?**

6. Which of the following statements is / are correct?

- Order scheduling assigns the production orders to the various work centers in the plant.
- Order scheduling monitors the status of the various orders in the plant.
- Order scheduling provides the documentation needed to process a production order through the factory.
- Order release provides the documentation needed to process a production order through the factory.
- Order progress module monitors the status of the various orders in the plant.

Answer:

- **Order scheduling assigns the production orders to the various work centers in the plant.**
 - **Order release provides the documentation needed to process a production order through the factory.**
 - **Order progress module monitors the status of the various orders in the plant.**
7. If a manufacturing system can accommodate changes in production scheduling, what flexibility test does it pass?
- New part
 - Error recovery
 - Part variety
 - Schedule change

Answer: Schedule change

8. What is the correct order of the three phases of a typical shop floor control system: A. Order progress B. Order release C. Order scheduling
- B, C, A
 - B, A, C
 - C, A, B
 - C, B, A
 - A, B, C

Answer: B, C, A

9. In production planning and control, what types of inventory are of greatest interest?
- Purchased components
 - Work in process
 - Finished products
 - Raw materials

Answer:

- **Purchased components**
 - **Work in process**
 - **Finished products**
 - **Raw materials**
10. The Just-in-Time production system operates successfully if:
- The setup time of the production line is small
 - The production is made in large batch sizes
 - Equipment with high reliability are used
 - The setup time of each equipment of the production system is large
 - The products are produced in small size batches

Answer:

- **The setup time of the production line is small**
- **Equipment with high reliability are used**
- **The products are produced in small size batches**